

Restriction Enzyme Digest and Ligation

This protocol outlines the digestion and ligation of a DNA insert into a plasmid. It begins by cutting both the source DNA and the plasmid with the same restriction enzymes, generating complementary ends that allow the fragments to align. The insert is then joined to the plasmid through a ligation reaction, which is facilitated using T4 DNA Ligase.

Reagents

✓	Name	Quantity	Location/Notes
☐	Restriction Endonucleases (e.g., NEB enzymes)	1 μ L each enzyme	Used in the 25 μ L restriction digest
☐	Restriction Digest Buffer (e.g., CutSmart Buffer)	3 μ L	Used in the 25 μ L restriction digest
☐	Plasmid DNA (target vector)	500 ng–1 μ g for digest; 50 ng for ligation	Digest uses more DNA; ligation specifically uses 50 ng vector
☐	Insert DNA (PCR product or synthesized DNA)	500 ng–1 μ g for digest; amount needed for chosen molar ratio in ligation	For ligation, usually based on 1:3 vector:insert for cohesive ends
☐	T4 DNA Ligase	1 μ L	Added last in the 20 μ L ligation reaction

☐	T4 DNA Ligase Buffer (10x)	2 μ L	Used in the 20 μ L ligation reaction
☐	Nuclease-free water	To bring digest to 25 μ L total; to bring ligation to 20 μ L total	Volume depends on how much DNA/enzyme/buffer is added

Equipment

✓	Name	Quantity	Location/Notes
☐	Thermal cycler or water bath		
☐	Microcentrifuge		
☐	Heat block (for enzyme inactivation)		
☐			
☐			

Consumables

✓	Name	Quantity	Location/Notes
☐	Pipettes and tips		
☐	Microcentrifuge tubes (1.5 mL)		
☐	PCR tubes		
☐			
☐			

Instructions

Restriction Digest Steps:

1. For the ligation step, first determine the appropriate molar ratio of the vector to insert. For cohesive-end ligations, a 1:3 ratio (vector:insert) is generally recommended, while blunt-end ligations may require a higher molar ratio such as 1:4 or 1:5 to improve efficiency.
2. Prepare a 25 μ L restriction digest reaction:
3. To begin the restriction digest, prepare a 25 μ L reaction mixture containing 500 ng – 1 μ g of plasmid DNA, 500 ng – 1 μ g of insert DNA, 1 μ L of each chosen restriction enzyme, 3 μ L of CutSmart™ Buffer, and nuclease-free water to adjust the final volume to 25 μ L.
4. Mix the components gently by pipetting up and down, ensuring proper enzyme contact with the DNA.
5. Incubate the reaction at 37°C for 1 hour, or at the temperature recommended for the selected restriction enzymes.
6. Once digestion is complete, heat inactivates the enzymes by incubating the mixture at 65–80°C for 20 minutes, depending on the enzyme's heat inactivation profile. This step prevents unwanted enzymatic activity during the subsequent ligation.

Ligation Steps:

1. For the ligation step, first determine the appropriate molar ratio of the vector to insert. For cohesive-end ligations, a 1:3 ratio (vector:insert) is generally recommended, while blunt-end ligations may require a higher molar ratio such as 1:4 or 1:5 to improve efficiency.

2. Prepare a 20 μL ligation reaction by combining 2 μL of T4 DNA Ligase Buffer, 50 ng of vector DNA, the appropriate amount of insert DNA (based on the chosen molar ratio), and nuclease-free water to bring the total volume to 20 μL .
3. Add 1 μL of T4 DNA Ligase last to ensure enzyme stability.
4. Mix the reaction gently by pipetting up and down.
5. Incubate the ligation mixture at 16°C overnight or for 1 hour at room temperature. For blunt-end ligations, an overnight incubation at 16°C or a 2-hour incubation at room temperature is recommended for improved efficiency.
6. Following incubation, heat inactivates the ligase at 65°C for 20 minutes to prevent unwanted activity. Chill the reaction on ice before proceeding to downstream steps such as transformation or storage.

Notes here

References

- Last year's protocol (Sterosaurus..not in the wiki but we did use this one)
- <https://static.igem.org/mediawiki/2018/a/a4/T--Tuebingen--RestrictionLigation.pdf>
- <https://www.neb.com/en-ca/protocols/optimizing-restriction-endonuclease-reactions>
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